

Table 1. RTC Accuracy Test Results

Time Interval	Laptop Time (hh:mm)	RTC Time (hh:mm)	RTC Accuracy (%)
Initial	18:39	18:39	100%
+5 minutes	18:44	18:44	100%
+10 minutes	18:49	18:49	100%
+15 minutes	18:54	18:54	100%
+20 minutes	18:59	18:59	100%
+25 minutes	19:04	19:04	100%
+30 minutes	19:09	19:09	100%
+35 minutes	19:14	19:14	100%
+40 minutes	19:19	19:19	100%
+45 minutes	19:24	19:24	100%
+50 minutes	19:29	19:29	100%
+55 minutes	19:34	19:34	100%
+60 minutes	19:39	19:39	100%

Table 2. Irrigation Schedule Data (Duration 60 Seconds)

Scheduled Start Time	Interval Between Valves	Valve 1 (hh:mm:ss)	Valve 2 (hh:mm:ss)	Valve 3 (hh:mm:ss)	Accuracy (%)
23:27	60 s	23:27:03	23:28:04	23:29:04	100%
23:33	60 s	23:33:04	23:34:04	23:35:04	100%
23:39	60 s	23:39:04	23:40:04	23:41:04	100%
23:49	60 s	23:49:03	23:50:04	23:51:04	100%
23:55	60 s	23:55:03	23:56:03	23:57:03	100%
23:58	60 s	23:58:03	23:59:04	00:00:04	100%
00:20	60 s	00:20:03	00:21:04	00:22:04	100%
00:24	60 s	00:24:03	00:25:04	00:26:04	100%
00:29	60 s	00:29:03	00:30:04	00:31:04	100%
00:34	60 s	00:34:03	00:35:04	00:36:04	100%

Table 3. Irrigation Schedule Data (Duration 120 Seconds)

Scheduled Start Time	Interval Between Valves	Valve 1 (hh:mm:ss)	Valve 2 (hh:mm:ss)	Valve 3 (hh:mm:ss)	Accuracy (%)
10:55	120 s	10:55:03	10:57:04	10:59:04	10:55
11:05	120 s	11:05:03	11:07:04	11:09:04	11:05
11:11	120 s	11:11:03	11:13:04	11:15:04	11:11
11:23	120 s	11:23:03	11:25:04	11:27:04	11:23
11:28	120 s	11:28:03	11:30:03	11:32:04	11:28
11:35	120 s	11:35:03	11:37:03	11:39:03	11:35
11:42	120 s	11:42:03	11:44:03	11:46:03	11:42
11:49	120 s	11:49:03	11:51:03	11:53:04	11:49
11:55	120 s	11:55:03	11:57:03	11:59:03	11:55
12:02	120 s	12:02:03	12:04:04	12:06:04	12:02

Table 4. Weather API Group

Weather Code	Thunderstorm	Weather Code	Drizzle	Weather Code	Rain
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200	Thunderstorm with light rain	300	Light Intensity Drizzle	500	Light Rain
201	Thunderstorm with rain	301	Drizzle	501	Moderate Rain
202	Thunderstorm with Heavy Rain	302	Heavy Intensity Drizzle	502	Heavy Intensity Rain
210	Light Thunderstorm	310	Light Intensity Drizzle Rain	503	Very Heavy Rain
211	Thunderstorm	311	Drizzle Rain	504	Extrem Rain
212	Heavy Thunderstorm	312	Heavy Intensity Drizzle Rain	511	Freezing Rain
221	Ragged Thunderstorm	313	Shower Rain and Drizzle	520	Light Intensity Shower Rain
230	Thunderstorm with Light Drizzle	314	Heavy Shower Rain and Drizzle	521	Shower Rain
231	Thunderstorm with Drizzle	321	Shower Drizzle	522	Heavy Intensity Shower Rain
232	Thunderstorm with Heavy Drizzle	801	Few Clouds	531	Ragged Shower Rain

```
// Cancel tasks if weather condition is not clear
if (weatherConditionCode != 801) {
    scheduler.cancelAllTasks();
    shouldRunSchedule = false;
    Serial.println("Weather is not clear.");
}
```

Fig. 3. Conditions Used for Simulation

```
Connecting to WiFi...
Connected to WiFi
RTC is running
System running...
HTTP code: 200
Raw JSON data:
{"coord":{"lon":110.4076,"lat":-6.9841},"weather":[{"id":801,"main":"Clouds","description":"few clouds","icon":"02d"}],"base":"stations","main":{"temp":29.97,"feels_like":32.39,"temp_min":29.97,"temp_max":29.97,"pressure":1011,"humidity":58,"visibility":6000,"wind":{"speed":6.69,"deg":100},"clouds":{"all":20},"dt":1718938917,"sys":{"type":1,"id":9362,"country":"ID","sunrise":1718923719,"sunset":1718965896},"timezone":25200,"id":1627896,"name":"Semarang","cod":200}}
Weather Description: few clouds
Weather Condition Code: 801
Scheduler Start from index: 0
```

Fig. 4. Serial Monitor When Conditions are Met